

● HARD

● PROBLEM-SOLVING AND DATA ANALYSIS

● ANSWER KEY

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# One-variable data: Distributions and measures of center and spread

# 01

10 answers · Rationale-rich · Teach from doc

**Q1****A****A. 0**

Choice A is correct. Since the frequencies of values less than the middle value, 3, are the same as the frequencies of the values greater than 3, the set of data has a symmetric distribution. When a set of data has a symmetric distribution, the mean and median values are equal. Therefore, the mean is 0 greater than the median.

Choices B, C, and D are incorrect and may result from misinterpreting the set of data.

**Q2****52**

The correct answer is 52. The mean of a data set is calculated by dividing the sum of the data values by the number of values. It's given that data set A consists of 10 values, 9 of which are shown. Let  $x$  represent the 10th data value in data set A, which isn't shown. The mean of data set A can be found using the expression  $\frac{43 + 45 + 44 + 43 + 38 + 39 + 40 + 46 + 40 + x}{10}$ , or  $\frac{378 + x}{10}$ . It's given that the mean of the 9 values shown is 42 and that the mean of all 10 numbers is greater than 42. Consequently, the 10th data value,  $x$ , is larger than 42. It's also given that the data values in data set A are positive integers less than 60. Thus,  $42 < x < 60$ . Finally, it's given that the mean of data set A is an integer. This means that the sum of the 10 data values,  $378 + x$ , is divisible by 10. Thus,  $378 + x$  must have a ones digit of 0. It follows that  $x$  must have a ones digit of 2. Since  $42 < x < 60$  and  $x$  has a ones digit of 2, the only possible value of  $x$  is 52. Since 52 is larger than any of the integers shown, the largest integer from data set A is 52.

**Q3****C****C. 32**

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Choice C is correct. If the mean score of 8 players is 14.5, then the total of all 8 scores is  $14.5 \times 8 = 116$ . If the mean of 7 scores is 12, then the total of all 7 scores is  $12 \times 7 = 84$ . Since the set of 7 scores was made by removing the highest score of the set of 8 scores, then the difference between the total of all 8 scores and the total of all 7 scores is equal to the removed score:  $116 - 84 = 32$ .

Choice A is incorrect because if 20 is removed from the group of 8 scores, then the mean score of the remaining 7 players is  $\frac{(14.5 \times 8) - 20}{7}$  is approximately 13.71, not 12. Choice B is incorrect because if 24 is removed from the group of 8 scores, then the mean score of the remaining 7 players is  $\frac{(14.5 \times 8) - 24}{7}$  is approximately 13.14, not 12. Choice D is incorrect because if 36 is removed from the group of 8 scores, then the mean score of the remaining 7 players is  $\frac{(14.5 \times 8) - 36}{7}$  or approximately 11.43, not 12.

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## Q4

B

- B. The mean of the new data set is greater than the mean of the original data set, and the medians of the two data sets are equal.
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Choice B is correct. The mean of a data set is the sum of the values in the data set divided by the number of values in the data set. The new data set consists of the weights of the 71 tortoises in the original data set and one additional weight, 39. Since the additional weight, 39, is greater than any of the values in the original data set, the mean of the new data set is greater than the mean of the original data set. If a data set contains an odd number of data values, the median is represented by the middle data value in the list when the data values are listed in ascending or descending order. Since the original data set consists of the weights of 71 tortoises and is in ascending order, the median of the original data set is represented by the middle value, or the 36th value. Based on the frequencies shown in the table, the 36th value in this data set is 17. If a data set contains an even number of data values, the median is between the two middle data values when the values are listed in ascending or descending order. Since the new data set consists of the weights of 72 tortoises, the median of the new data set is between the 36th and 37th data values when the values are arranged in ascending order. To keep the data in ascending order, the additional value of 39 would be placed at the bottom of the frequency table with a frequency of 1. Therefore, based on the frequencies in the table, the 36th and 37th values in the new data set are both 17. It follows that the median of the new data set is 17, which is the same as the median of the original data set. Therefore, the mean of the new data set is greater than the mean of the original data set, and the medians of the two data sets are equal.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

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**Q5****44**

The correct answer is 44. The mean of a data set is computed by dividing the sum of the values in the data set by the number of values in the data set. It's given that data set A consists of the heights of 75 buildings and has a mean of 32 meters. This can be represented by the equation  $\frac{x}{75} = 32$ , where  $x$  represents the sum of the heights of the buildings, in meters, in data set A. Multiplying both sides of this equation by 75 yields  $x = 75(32)$ , or  $x = 2,400$  meters. Therefore, the sum of the heights of the buildings in data set A is 2,400 meters. It's also given that data set B consists of the heights of 50 buildings and has a mean of 62 meters. This can be represented by the equation  $\frac{y}{50} = 62$ , where  $y$  represents the sum of the heights of the buildings, in meters, in data set B. Multiplying both sides of this equation by 50 yields  $y = 50(62)$ , or  $y = 3,100$  meters. Therefore, the sum of the heights of the buildings in data set B is 3,100 meters. Since it's given that data set C consists of the heights of the 125 buildings from data sets A and B, it follows that the mean of data set C is the sum of the heights of the buildings, in meters, in data sets A and B divided by the number of buildings represented in data sets A and B, or  $\frac{2,400 + 3,100}{125}$ , which is equivalent to 44 meters. Therefore, the mean, in meters, of data set C is 44.

**Q6****D**

D. Neither I nor II

Choice D is correct. It's given that data set F consists of 55 integers between 170 and 290 and data set G consists of all the integers in data set F as well as the integer 10. Since the integer 10 is less than all the integers in data set F, the mean of data set G must be less than the mean of data set F. Thus, the mean of data set F isn't less than the mean of data set G. When a data set is in ascending order, the median is between the two middle values when there is an even number of values and the median is the middle value when there is an odd number of values. It follows that the median of data set F is either greater than or equal to the median of data set G. Therefore, the median of data set F isn't less than the median of data set G. Thus, neither the mean nor the median must be less for data set F than for data set G.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

**Q7**

**D**

D. Standard deviation

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Choice D is correct. When a data set has an odd number of elements, the median can be found by ordering the values from least to greatest and determining the middle value. Out of the 27 different numbers in this data set, 13 numbers are below the median, one number is exactly 33, and 13 numbers are above the median. When 7 is subtracted from each number below the median and added to each number above the median, the data spread out from the median. Since the median of this data set, 33, is equivalent to the mean of the data set, the data also spread out from the mean. Since standard deviation is a measure of how spread out the data are from the mean, a greater spread from the mean indicates an increased standard deviation.

Choice A is incorrect. All the numbers less than the median decrease and all the numbers greater than the median increase, but the median itself doesn't change. Choices B and C are incorrect. The mean of a data set is found by dividing the sum of the values by the number of values. The net change from subtracting 7 from 13 numbers and adding 7 to 13 numbers is zero. Therefore, neither the mean nor the sum of the numbers changes.

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**Q8**

**C**

- C. The median of data set B is greater than the median of data set A, and the range of data set B is equal to the range of data set A.
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Choice C is correct. The median of a data set with an odd number of values, in ascending or descending order, is the middle value of the data set, and the range of a data set is the positive difference between the maximum and minimum values in the data set. Since the dot plot shown gives the values in data set A in ascending order and there are 15 values in the data set, the eighth value in data set A, 23, is the median. The maximum value in data set A is 26 and the minimum value is 22, so the range of data set A is  $26 - 22$ , or 4. It's given that data set B is created by adding 56 to each of the values in data set A. Increasing each of the 15 values in data set A by 56 will also increase its median value by 56 making the median of data set B 79.

Increasing each value of data set A by 56 does not change the range, since the maximum value of data set B is  $26 + 56$ , or 82, and the minimum value is  $22 + 56$ , or 78, making the range of data set B  $82 - 78$ , or 4. Therefore, the median of data set B is greater than the median of data set A, and the range of data set B is equal to the range of data set A.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

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**Q9****D**

D. 15.25

Choice D is correct. The mean amount of time that the 20 employees worked for the company is 6.7 years. This means that the total number of years all 20 employees worked for the company is  $(6.7)(20) = 134$  years. After the employee left, the mean amount of time that the remaining 19 employees worked for the company is 6.25 years. Therefore, the total number of years all 19 employees worked for the company is  $(6.25)(19) = 118.75$  years. It follows that the number of years that the employee who left had worked for the company is  $134 - 118.75 = 15.25$  years.

Choice A is incorrect; this is the change in the mean, which isn't the same as the amount of time worked by the employee who left. Choice B is incorrect and likely results from making the assumption that there were still 20 employees, rather than 19, at the company after the employee left and then subtracting the original mean of 6.7 from that result. Choice C is incorrect and likely results from making the assumption that there were still 20 employees, rather than 19, at the company after the employee left.

**Q10****2**

The correct answer is 2.6. Since the mean of a set of numbers can be found by adding the numbers together and dividing by how many numbers there are in the set, the mean mass, in kilograms, of the rocks Andrew collected is  $\frac{2.4+2.5+3.6+3.1+2.5+2.7}{6} = \frac{16.8}{6}$ , or 2.8.

Since the mean mass of the rocks Maria collected is 0.1 kilogram greater than the mean mass of rocks Andrew collected, the mean mass of the rocks Maria collected is  $2.8+0.1=2.9$

kilograms. The value of  $x$  can be found by writing an equation for finding the mean:

$\frac{x+3.1+2.7+2.9+3.3+2.8}{6} = 2.9$ . Solving this equation gives  $x = 2.6$ . Note that 2.6 and

$\frac{13}{5}$  are examples of ways to enter a correct answer.

Notes



Accept equivalent forms (e.g.  $0.25 \equiv 1/4$ ). For grid-in items, decimal and fraction answers are both valid.

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